

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: C Programming for Problem Solving (CPPS) | Course Code: 25EN1114 | Credits: 4

Exam Session: Mid Semester Exam (CIA-1) March 2026 (Max Marks: 40, Duration: 15 min) | Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Write an algorithm, draw a flowchart, and develop a complete C program to compute electricity bill charges based on slab rates: 0-100 units @ Rs 1.50/unit, 101-200 units @ Rs 2.00/unit, >200 units @ Rs 3.00/unit, with a minimum charge of Rs 100 and a 15% surcharge if the bill exceeds Rs 400. [8 Marks]

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Q2. Demonstrate the step-by-step trace of Selection Sort algorithm on the array [64, 25, 12, 22, 11] in ascending order, displaying intermediate array state after each pass. [8 Marks]

MODULE 2

Q3. Explain Dynamic Memory Allocation in C. Compare malloc(), calloc(), realloc(), and free() with their syntax, return types, and memory layout diagrams. [8 Marks]

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Q4. Develop a C program using nested structures to store student information (Name, Roll No, Date of Birth {dd, mm, yyyy}, and Marks in 3 subjects). Compute total marks, average, and print grade. [8 Marks]